

The Wheel: UX Processes, Lifecycles, Methods, and Techniques

Highlights

- The need for a process
- Basic process components for UX
 - UX design lifecycle process
 - The Wheel: a UX design lifecycle template
 - UX lifecycle activities
 - UX design methods
 - UX design techniques
 - The UX technique toolbox
- Fundamental UX lifecycle activities
 - Understand needs
 - Design solutions
 - Prototype candidates
 - Evaluate UX
- UX design techniques as life skills
- Choosing and appropriating UX methods

2.1 INTRODUCTION

2.1.1 This Is Not About Software

This book is all about UX design, but not at all about software. What we produce is a UX design, usually represented as a wireframe specification or prototype (see [Section 18.3](#)). These designs will be realized in software by developers (also called software engineers or programmers), using a corresponding software engineering lifecycle (see [Section 26.3](#)).

2.1.2 The Need for Process

Long ago, software engineering folks recognized that having a process is necessary for developing complex systems, and they invested enormous resources (Paulk et al., 1993) into defining, verifying, and following a process. Similarly, on the UX side, the need for a guiding process was recognized early (Whiteside & Wixon, 1985).

2.1.3 What Do You Get by Having a Process?

Process is a guiding structure. A process is a guiding structure that helps both novices and experts deal with the activities and complex details of a project. A process enforces a systematic approach, bringing order to what could be very chaotic, especially within a large and complex project. Without a process, designers and developers must make it up as they go.

A process acts as a framework to ensure designers, especially novices, are on track to a quality product and on the path to becoming experts. For experts, a process acts as a checklist to make sure they do not miss any important aspects of the problem in the heat of productivity. It helps designers answer questions, such as “Where are we now?” and “What can/should we do next?” Without a process, you are limited by your own experience.

Process offers reliability and consistency. A documented process offers a way to use the same basic approach from project to project and from team member to another. Pressure for fast turnaround (and the risk of resulting poor quality) is moderated by required process steps and checks.

Process provides scaffolding for learning. Design is all about learning. A process provides a fabric on which you can build a knowledge base of what you have learned, applying organizational memory from similar previous efforts to incorporate lessons learned in the past. This fabric, in turn, helps train new or novice designers in the ways of UX at that organization or in the discipline at large.

Process provides a shared conception of what you are doing. A documented process lets everyone know how a product or system (software plus UX) is being developed. Process also helps your team coordinate and communicate by externalizing the state of development for observation, measurement, analysis, and control—otherwise, communication among the project roles about what they are doing is difficult because they lack a shared concept of those roles.

2.1.4 A Hierarchy of Terms

To distinguish the levels among descriptions of ways things can be done in UX design, we end up with a kind of hierarchy of terminology:

- Process, or UX lifecycle process
- UX lifecycle activities and sub-activities
- UX methods
- UX techniques

These terms, despite their common appearance in the literature, are usually vague and ill-defined within the discipline. We have chosen to shade the definitions of these closely related terms to reflect the meanings we think are most commonly understood within this loose kind of hierarchical relationship, examples of which are shown in [Table 2.1](#).

2.2 THE WHEEL: A MODEL OF THE UX DESIGN PROCESS

2.2.1 UX Design Lifecycle Process

A UX design lifecycle is just what it says: It is a cycle of the life of a UX design, from inception to deployment and beyond. It is usually represented in the form of a flow chart diagram connecting the flow of activities in a sequence over time. The representation in [Fig. 2.1](#) is of the overall UX design lifecycle process, the focus of this book. The boxes are lifecycle activities, high-level things done during a UX design lifecycle.

Fine distinctions are unnecessary here, so we use the terms *process*, *lifecycle*, and *lifecycle process* more or less interchangeably.

Table 2.1
Informal hierarchy of process, methods, and techniques with simple examples

Term	Example(s)
Lifecycle process	Traditional waterfall process (see Section 4.2)
Lifecycle activity	Understand needs (see Part 2)
Sub-activity	UX research data gathering (see Chapter 6)
Method	Client visits
Technique	Interviews, observations

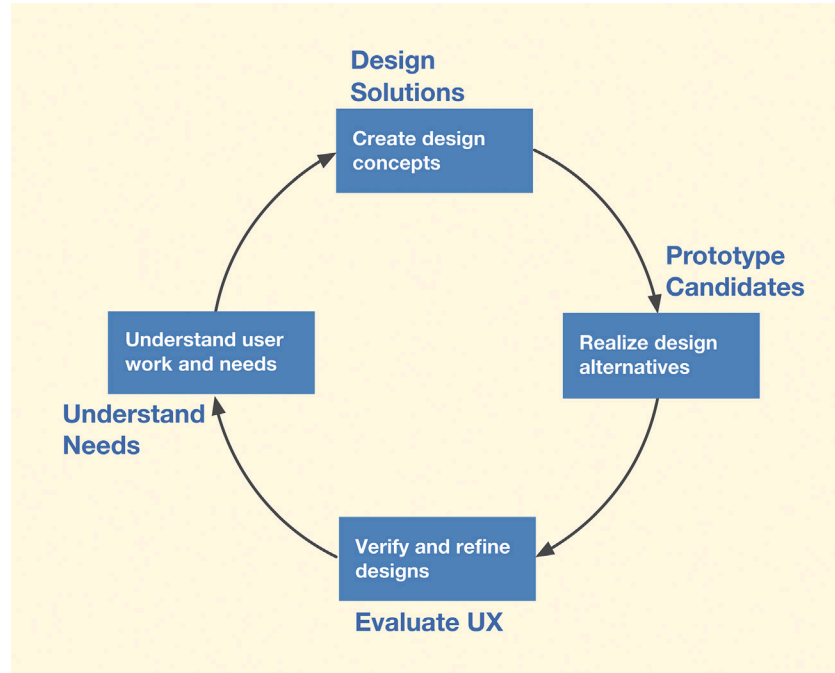


Fig. 2.1

The basic UX design lifecycle process in terms of elemental lifecycle activities.

2.2.2 The Wheel as a UX Design Lifecycle Template

If we expand the abstract cycle of Fig. 2.1 a bit to include feedback and iteration, we get a kind of UX design lifecycle template (Fig. 2.2). We call this The Wheel because it goes around in circles, and with each rotation it brings you closer to your destination.

This basic picture is the blueprint for the process common to almost any kind of design; it applies whether the design scope is only a small piece of a product/system or the whole system.

2.2.3 UX Design Lifecycle Activities

Lifecycle activities are the high-level things you do during a lifecycle (note boxes in Figs. 2.1 and 2.2). Much of this book is about these four basic UX design lifecycle activities:

1. **Understand needs**, to understand users, work¹ practice, usage, the subject-matter domain, and, ultimately, needs for the design (see Part 2)

¹When we use the term *work*, we intend it to broadly signify anything people/users do while using products or systems. See Section 6.2.

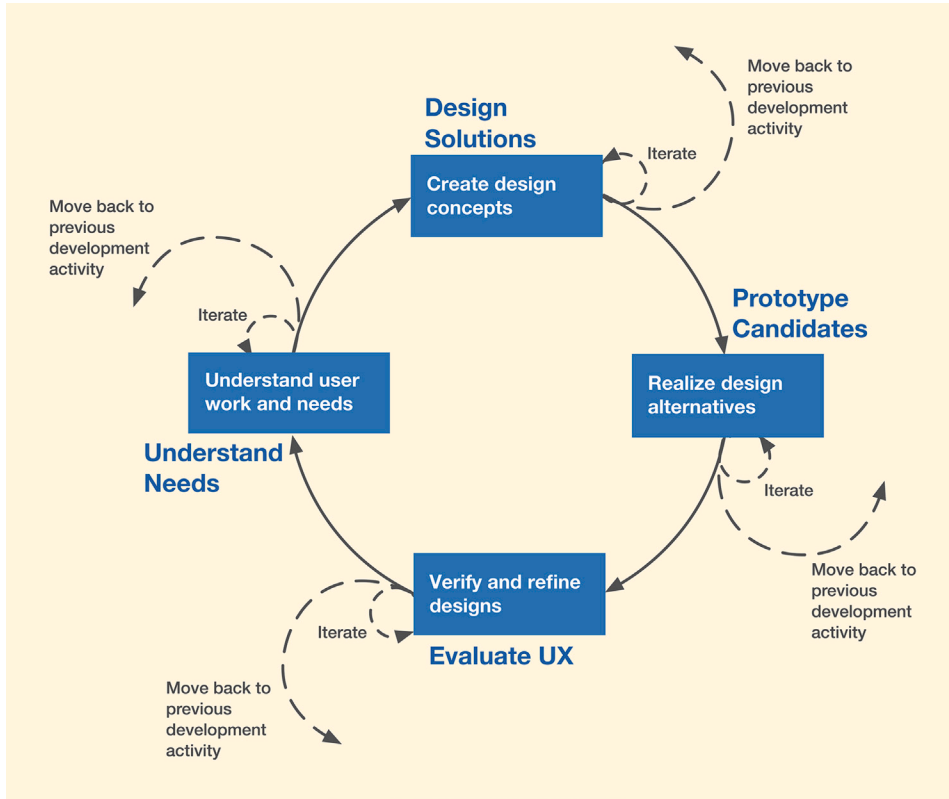


Fig. 2.2

The Wheel is a lifecycle template illustrating the process part of this book.

2. **Design solutions**, to create designs as solutions to satisfy user needs (see [Part 3](#))
3. **Prototype candidates** (of solutions), to realize and envision promising design candidates (see [Part 4](#))
4. **Evaluate UX**, to verify and refine designs with respect to the user experience they afford (see [Part 5](#))

The depiction of UX lifecycle activities in distinct boxes is a convenient way to highlight each activity for discussion and for mapping to chapters in this book. In practice, however, these activities do not have such clear-cut boundaries; there can be significant intertwining and overlap (see [Chapter 5](#)).

2.2.4 Lifecycle Sub-Activities

Each lifecycle activity is substantial enough to be described in terms of its own set of sub-activities. Lifecycle sub-activities are the things you do during a single lifecycle activity.

Example sub-activities for the understand-needs activity (one of the four basic UX design lifecycle activities; see earlier) understand-needs include:

- Data gathering (see [Chapter 6](#)): Interview and observe users at work or otherwise research that work and gather data about work practice, users, usage, and needs.
- Data analysis (see [Chapter 7](#)): Distill and organize UX research data.
- Data modeling (see [Chapters 8 and 9](#)): Create representations of user characteristics, information flow, tasks, and work environments (for understanding, collaboration, sharing, archival, rehearsal, immersion).
- User stories and requirements (see [Chapter 10](#)).

2.2.5 UX Design Methods

We use the term *method* as a generic word referring to a general approach to carrying out any activity. An example of a method for the understand-needs lifecycle activity is UX research (see [Part 2](#)).

The goals of an activity or sub-activity can be accomplished through one or more methods. For example, methods abound for doing the UX evaluation activity, from lightweight and rapid to thorough and laborious, from full empirical studies to quick and dirty inspections (see [Part 5](#)).

2.2.6 UX Design Techniques

In addition to the UX processes and methods described in this chapter, there exist a number of other UX techniques. In our usage, a UX technique is a procedure or a specific detailed practice you can use to perform a step within an activity, sub-activity, or method. A given UX design technique could be useful in a variety of different lifecycle activities and is not tied to a specific UX method. Examples of UX techniques for gathering data within a method for UX research are:

- Interviewing (of users)
- Observation (of users at work)

2.2.6.1 UX techniques are also life skills

Because these techniques are also used for problem-solving in our everyday lives, we think of them as life skills—basic generic skills for solving problems, helping designers and non-designers thrive as human beings. Most should seem familiar because you will have encountered them or used them in other

contexts. For example, we utilize many of these techniques when navigating life events, such as renting or buying a house:

- **Observation:** We explore different houses to learn about the area, the noise levels, traffic patterns during different times of the day, where and how sunlight enters the place, how interactions with neighbors work, among other things.
- **Note taking:** We take notes to capture what we observe for later discussion or reflection, particularly if we are considering multiple housing options.
- **Sketching:** We may sketch the layout of the house and annotate it with comments and insights.
- **Modeling:** For the more designerly among us, we may even model typical workflows, such as bringing groceries home from the store through unloading them into the pantry and refrigerator.
- **Ideation:** If we are considering building a house instead, we may brainstorm ideas for various features of the house and even prototype a few concepts.

To give you a bit of upfront familiarity for when you encounter applications of these techniques in later chapters, we preview some common UX techniques here in our UX technique toolbox.

2.3 THE UX TECHNIQUE TOOLBOX

Numerous UX design techniques exist, and more are being described in the literature and used in practice over time (Martin & Hanington, 2012). It is not practical for us to cover all these in this book. We have included a few UX techniques in this toolbox that we have found to be versatile in our practice and experience.

2.3.1 Techniques for Research and Exploration

2.3.1.1 Observation

Observation is the practice of witnessing an ongoing activity with the objective of understanding underlying phenomena. Things to look for include exceptions, surprises, generalities, patterns, workflows, sequencing, what works and what does not work, problems and barriers, and how people react to problems (or if they do). Observation provides the inputs for reasoning and deduction ([Section 2.3.3.3](#)). The ability to observe effectively can be elusive, and it can take practice to develop the skill to observe. UX professionals must train themselves to recognize when they are seeing something important to avoid letting it slip away because it did not register.

See [Section 6.7](#) for more details about how we use observation to learn about users' work, problems and breakdowns, and opportunities for improvement. See [Section 22.3](#) for how we use observation in UX evaluation where we focus on user behavior while interacting with a design to learn about defects and quality of the user experience.

2.3.1.2 *Note taking*

Note taking is the practice of efficiently capturing descriptions of observations and other forms of qualitative data. Techniques for note taking include making handwritten notes, typing notes, and recording the essence on audio or video. Regardless of how you do it, note taking should be done as an almost subconscious activity without distracting your cognitive processes from the observation activity.

To be efficient, you must apply abstraction ([Section 2.3.3.2](#)) during note taking to capture the essential points while keeping the verbiage to a minimum. Notes can include sketches and/or models, analogies, or any other descriptive mechanism, bringing additional techniques into the mix.

See [Section 6.7.8](#) for more details about note taking in UX research. See [Section 22.3.1](#) for using the note-taking technique while observing a UX evaluation session.

2.3.1.3 *Sketching*

Sketching in UX is the practice of making simple pictures and diagrams depicting the essence of problems and solutions. Sketching is a way to externalize analysis and exploration of objects, their relationships, and an emerging understanding of the problems and solutions. The most important point about sketching is that it is not about art or aesthetics. It is about communication of ideas.

A sketch is a kind of prototype. It uses an abstract representation, highlighting the salient features to aid visualization. Sketching can boost creativity by connecting the embodiment of hand-eye activity (Graves, 2012) with cognition in the brain. Sketching must always occur with ideation. As Buxton (2007b) says, "If you are doing design, you are sketching." See [Section 13.4](#) for a description of how sketching is used as an integral part of generative design.

Sketching is a life skill with wide range of applications in everyday life. Reorganizing furniture in your home? First make a sketch of the envisioned configuration or layout. Substantiate it with a model of your workflows for that room and immerse yourself in the usage context of that room.

2.3.2 Techniques for Organizing Data and Ideas

Techniques for organizing data and ideas are used to sort items by category, usually a hierarchy of categories and sub-categories. This sorting has the effect of collecting related notes together in shared categories and of illustrating the hierarchical or other types of relationships among categories to underscore important issues and afford insights useful for design.

2.3.2.1 Hierarchical lists

For small amounts of data and simple categories, hierarchical lists are an economical approach to organization. The highest level of categories is represented at the top of the hierarchy. Next-level sub-categories are indented under their corresponding parent categories. Fig. 2.3 shows the hierarchical structure for an overly simplified example.

The corresponding hierarchical list will look something like this:

```

Veterinary business
  Clientele
    Large animals
      Horses
        (data notes about horses)
  
```

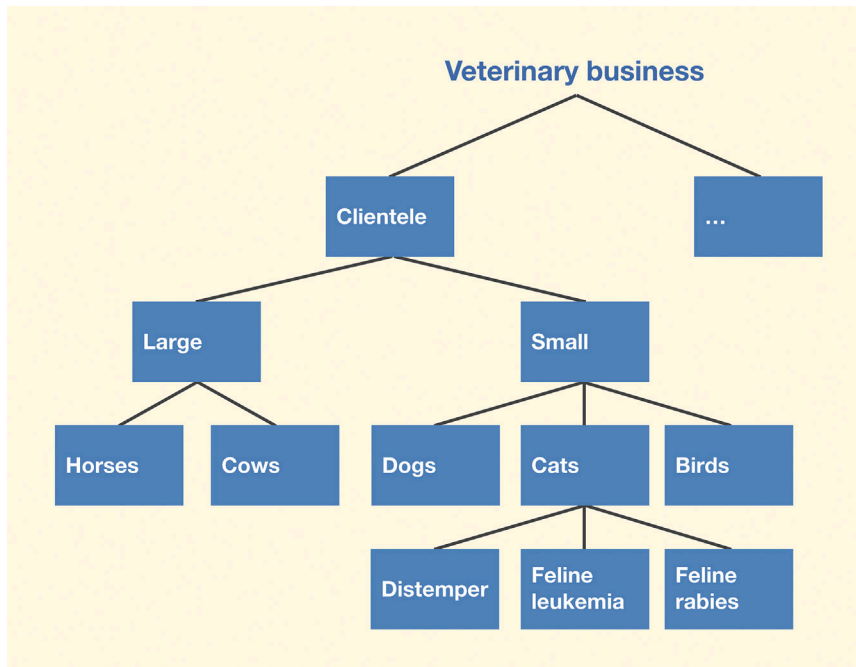


Fig. 2.3
Example of a structured
hierarchical list.

- Cows
 - (data notes about cows)
- Small animals
 - Dogs
 - (data notes about dogs)
 - Cats
 - (data notes about cats)
 - Distemper
 - (data notes about distemper)
 - Feline leukemia
 - (data notes about feline leukemia)
 - Feline rabies
 - (data notes about feline rabies)
- Birds

2.3.2.2 Card sorting

Card sorting is a technique for ordering cards annotated with concepts into categories based on shared characteristics. Card sorting is usually limited to small amounts of data (e.g., to organize menu structures in desktop applications). Designers list all the actions the system should support, and each action is printed on a card. Users or other stakeholders are then asked to organize them into appropriately related groups.

2.3.2.3 Concept/mind mapping

Mind mapping² or concept mapping³ is a technique for drawing labeled images centered on concepts and relationships among them as a way to explore new ideas or concepts relative to other known ideas and concepts. The flow model (Section 8.4) is an example of a special type of concept map that depicts how various things of interest to design move through a work practice.

2.3.2.4 Affinity diagramming

Affinity diagrams are used to organize large amounts of data, grouping data hierarchically by relatedness (affinity) and common attributes (Section 7.5.1). It is a technique for organizing data collected, for example, in UX research or in UX evaluation, identifying themes and relationships to understand the whole corpus of data. Affinity diagramming, especially when adapted as work

²https://en.wikipedia.org/wiki/Mind_map

³https://en.wikipedia.org/wiki/Concept_map

activity affinity diagrams (WAADs) for UX research organization, is a powerful tool for organizing data (work activity notes) by category. For large amounts of UX research data notes, we recommend a WAAD. See [Section 7.5](#) for a detailed description of building a WAAD.

2.3.3 Techniques for Understanding and Sense Making

2.3.3.1 Immersion

Immersion is a form of deep thought and analysis of the problem at hand—to live within the context of a problem to discover and understand its various dimensions, attributes, constraints, and other properties.

Immersion is about surrounding yourself in your UX war room, so to speak (see UX design studio in [Section 5.2](#)), with your design artifacts (posters, notes, sketches, photos, diagrams, quotations, goal statements). You close yourself off from outside distractions, and everything you see acts as a kind of cognitive scaffolding and a catalyst that helps spawn design ideas.

Consider this example of immersion in a non-UX setting. When a police detective is trying to solve a difficult crime, she might isolate herself in a room away from distractions. She becomes steeped in the problem and surrounds herself with artifacts, such as photos and sketches of the crime scene, police and witness reports, and a timeline on the wall. She studies the problem so intensely that she becomes the perpetrator, in a sense.

2.3.3.2 Abstraction

Abstraction is a powerful technique based on removing detail irrelevant to a given objective or focus of discourse. Abstraction is one method for separating the wheat from the chaff and boiling things down to their essence.

“Abstraction is considered to be a key skill underlying most activities in computer science...and software engineering” (Hazzan & Kramer, 2016). The result is a clearer picture of what is important without the distraction of extraneous matter. Abstraction also entails the ability to generalize from an example to a more general case or principle.

Consider the case of building a house, where as a designer (architect) you are interviewing users (potential residents) of the house in question. Different users mention different needs:

- User A: I would like to have a fan in the kitchen to clear any cooking smells.
- User B: I like to open my windows to the study and let in fresh air.
- User C: I do a lot of gluing in my workshop and need to have large windows and doors so the chemical smells do not overwhelm me.

All these specific instances can be abstracted under a generic idea of ventilation for the house that will lead to a design that solves all the individual problems in the list.

2.3.3.3 Reasoning and deduction

Reasoning and deduction constitute a long-standing practice of applying logic to process observed facts, fit them together, and arrive at a logical conclusion. Reasoning and deduction are a way of synthesizing new facts through the use of logic as applied to existing facts. We will see examples of this in [Chapter 10](#) where we deduce system requirements from UX research data.

2.3.3.4 Framing and reframing

Framing and reframing is a conceptualization of a design problem in a particular perspective based on a select few prominent features. Framing reduces complexity and helps focus on the most important dimensions of the problem. For example, confusion surrounding a collection of disparate kinds of data might be cleared up by framing it within a spreadsheet.

Framing builds a perspective that structures the problem and highlights the aspects you will explore. A framing is a pattern or a particular theme from which we view everything during the process of finding solutions. It is a specific context in which we can ask, “What if?” and “Why don’t we do this?”

Here is a non-UX example of how different framings lead to different solutions in any kind of problem-solving in almost any setting. Suppose you have occasional flooding of fields along the floodplains of a river near a town. If you frame it as a problem of too much water coming down the river, then it will lead you to focus on design solutions in the form of dams or other ways to control the flow into the valley. Alternatively, if you frame it as a problem of the river overflowing its banks, then you might turn to a system of dikes to keep the water within the banks. Or, if you frame it as a natural occurrence to work with instead of fighting against, then you will require landowners to build homes only above the floodplain.

Reframing is a transformation of perspective. Mathematicians practice this skill when they switch coordinate systems to study a phenomenon for which a new coordinate system is more suitable. For example, early 20th-century electrical engineers moved from traditional cartesian coordinates to a cylindrical coordinate system to study electric fields within cylindrical vacuum tubes. See [Section 13.3.5.1](#) for an example of how a simple reframing solved a big industrial engineering problem.

2.3.3.5 Synthesis

Synthesis involves combining things (e.g., data) you already have or know to create something new. Synthesis is closely associated with creativity and innovation, meaning to create new ideas and understandings that were not present in the original data. That is why the early stages of creative design are called generative (see [Chapter 13](#)); they generate new design ideas. More generally, synthesis is a way of visualizing the full story in a situation that requires processing multiple inputs and drawing conclusions.

One of the best examples of the importance of synthesis skills outside UX is the way they are used by police detectives when they put together disparate clues, synthesizing observations, interviews, note taking, framing, and reasoning/deduction as they build a complete case to figure out who is the perpetrator.

Organizing UX research data ([Section 7.4](#)) and building data models (see [Chapters 8 and 9](#)) are good examples of applying synthesis to raw UX research data to produce new ideas, insights, and realizations that did not exist in the original data. Other tools in the UX technique toolbox, such as framing and reframing, along with envisioning, work hand in hand with synthesis. These techniques are all enhanced through immersion, putting ourselves in direct relationship with the data.

Much of our thinking about design synthesis has been strongly influenced by Kolko (2007), who stated, “Contextual research, competitive research, or standard academic research is used to gather data, but that data alone does not indicate the potential for strategic design thinking; the data must be analyzed, organized, and understood. . . . It is a messy phase with strategic implications and strategic importance.” Like most other UX techniques, synthesis has a broad range of applications. See [Sections 7.3 and 13.1.3](#) for more.

2.3.3.6 Modeling

Modeling is the practice of representing complex and abstract phenomena along particular dimensions to simplify and aid understanding. A special kind of abstraction, modeling is a way to explain or categorize aspects of the problem space and to identify and represent objects, relationships, actions, operations, variables, and dependencies within that space.

As an example of modeling as a life skill, consider the research you do to compare possibilities before buying a car. You will encounter all kinds of information about cars. Your model might include dimensions such as

cost, style (convertible, SUV, sedan), and mechanical specifications (MPG, horsepower, torque, all-wheel drive vs. front-wheel drive).

2.3.4 Techniques for Generating, Envisioning, and Refining

2.3.4.1 Brainstorming

Brainstorming is the practice of interactive group discussion for exploring and prompting different ideas, problems, and solutions—more specifically:

- It is most effective when done as a group activity where each person's inputs and discussion stimulate others, triggering or inspiring additional thoughts.
- It is a major skill in the design-solutions lifecycle activity to highlight different perspectives and generate different framings of a phenomenon or a problem.
- It can be used in the evaluate-UX lifecycle activity to create solutions to identified UX problems.
- It can be used in any situation where the problem is open-ended (e.g., Who are potential users of this system? Where can we find participants for evaluation?).

More about brainstorming as part of ideation, sketching, and critiquing in UX design can be found in [Chapter 13](#) on generative design.

2.3.4.2 Prototyping for envisioning

Prototyping is the practice of producing or building a model or mockup of a design that can be manipulated and used at some level to manifest or simulate a user experience, which can be evaluated.

Prototyping extends the idea of sketching. As the main output of UX design, a prototype is a tangible platform for envisioning and evaluating efficacy of a design as a problem solution. See [Chapter 18](#) for much more about prototyping.

2.3.5 Techniques for Communicating

2.3.5.1 Storytelling

Storytelling is the practice of using narrative to explain aspects of a phenomenon or design with the objective of immersing the audience in the phenomenon.

A good example of storytelling is its use in advertising. It can be more compelling to tell stories of people who use a product and who get pleasure and/or utility from it in their lives than to simply list advantages of the product.

2.3.5.2 Externalizing insights

Visually posting data organization and other results (e.g., in the UX studio or war room) lets designers off-load their human memories and gives a large visual and mental bandwidth for literally walking through the data. Spatial/visual representation reduces the mess of data (Kolko, 2010), and posting provides a platform or scaffolding to support immersion.

2.4 CHOOSING UX PROCESSES, METHODS, AND TECHNIQUES

2.4.1 UX Techniques Are Used in Combination

When used in UX design or as life skills, these techniques are usually combined within methods. For example, a news reporter must combine skills to dig out and produce news stories to report, including observation, note taking, storytelling, and framing.

Within any given project, you have to choose UX processes, methods, and techniques for the UX lifecycle activities and sub-activities.

2.4.2 The UX Lifecycle Process Choice

The UX lifecycle process choice is made at the highest level. The way things have worked out in the world has had a large influence on that choice. The software engineering (SE) world has adopted an agile lifecycle process almost universally, and we, in UX, are making the same choice to keep pace with our SE partners. As in SE, an agile UX process is one in which you manage change during the process by delivering UX designs in small chunks.

Much of this book can apply to non-agile UX lifecycle processes, but our intent is to focus on agile UX. As we will soon see, that can mean many things depending on multiple factors and where you are in the overall process. In [Chapters 4 and 26](#), we will elaborate on what this means in practice.

2.4.3 Appropriating Methods and Techniques

There is an abundance of UX design methods. “HCI is awash in methods and the theories that underlie them” (Harrison, Back, & Tatar, 2006). So, how can you make sense of what you need for your project?

The usual practice in a book like this, or in courses on UX design, is for authors or instructors to take a method-oriented approach to lay out their favorite UX methods and proceed to show why that is the way to do UX design. In this edition, we think it makes more sense to begin with the idea of a design situation, the parameters of the project, user communities, and tasks

intended, and then talk about how to match it by appropriating methods and techniques that can achieve the goals and expected outcomes.

This idea of appropriating design methods comes from Harrison and Tatar (2011) and is an excellent example of synthesis. It simply means that you take standard methods you have learned about, and modify, adapt, and combine them, so they become your own in specific design situations.

2.4.3.1 Design situations: Dependencies that govern lifecycle activity, method, and technique choices

Harrison and Tatar (2011) describe a design situation as the circumstances under which a design method will be appropriated and applied. A good umbrella term is *design situation* because it includes the target product or system as well as the project and all of its context, including the market, the demand for rigor, pressure from a client or management for rapid turnaround, subject-matter domain complexity, and project team capabilities, skills, and experience.

2.4.3.2 Choosing methods and techniques

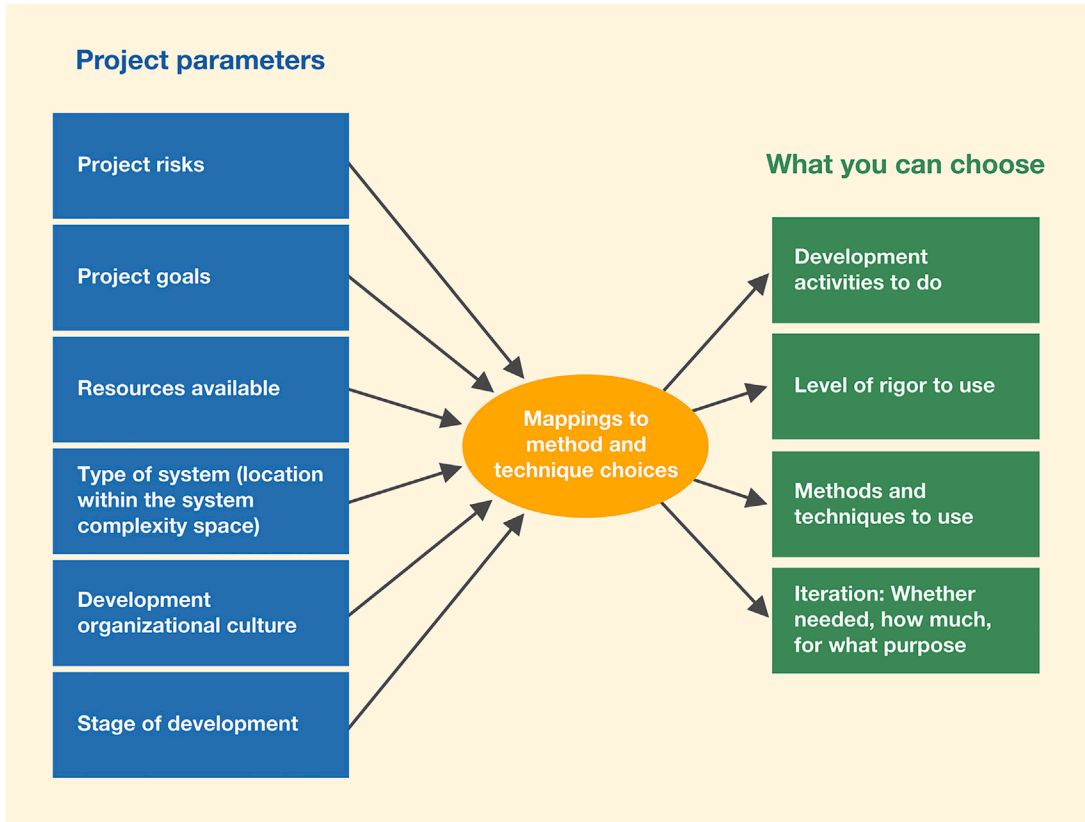
As you undertake a lifecycle activity, you will have a set of methods and techniques at hand. For example, suppose you need to choose a method to carry out the understand-needs lifecycle activity. One of the most popular of such methods is called UX research, a method for interviewing and observing real users to understand their work activities (see [Part 2](#)). Other methods to understand user needs, which are less popular but still applicable in some specific design situations, are focus groups and user surveys.

2.4.3.3 Early method and technique choices constrain later ones

Earlier choices of methods and techniques can constrain later choices by suggesting, eliminating, or dictating appropriate methods and techniques for subsequent choices. For example, methods and techniques used for data analysis in a given situation will depend on what kind of data you have and how the data were collected.

2.4.3.4 Mapping project parameters to lifecycle activity, method, and technique choices

To summarize, in [Fig. 2.4](#) we show the mapping from project parameters to possible choices of UX methods. While there are some general guidelines



*Fig. 2.4
Mapping
project
parameters
to agile
UX method
choices.*

for making these mapping choices, fine-tuning is the job of project teams, especially in consultation with the product owner (the person responsible for success of the product and for pursuing business goals; see [Section 5.4.5](#)). Much of it is intuitive and straightforward.

2.5 SEGUE

In this chapter we introduced The Wheel as a structure to frame UX processes, lifecycles, methods, and techniques, and how to map them to real-world project contexts. In the next chapter, we dig deeper into two aspects of the project development space: scope and rigor.